As they move away in both directions, newly synthesized complementary nucleotides are paired with the existing nucleotides on the parent strand and covalently bonded together by **DNA polymerase**. Formation of new strand requires a primer (a short stretch of RNA)for initiation. The primer produces a 3'-OH end on the sequence of ribonucleotides, to which deoxy ribonucleotides are added. The RNA primer is ultimately removed leaving a gap in the newly synthesized DNA strand. It is removed from 5' end one by one by the exonuclease activity of DNA polymerase. Finally, when all the nucleotides are in position, gaps are sealed by the enzyme **DNA ligase**.

At the point of origin of replication, the helicases and topoisomerases (DNA gyrase) unwind and pull apart the strands, forming a Y-Shaped structure called the **replication fork**. There are two replication forks at each origin. The two strands of a DNA helix have an antiparallel orientation. The enzyme DNA polymerase can only catalyse the addition of a nucleotide to the new strands in the 5'⟶3' direction, as it can only add nucleotides to the 3' carbon position.

## 5.9 Transcription

Francis Crick proposed the **Central dogma** of protein synthesis in molecular biology states that genetic information flows as follows:




The process of copying genetic information from one strand of DNA into RNA is termed **transcription**. This process takes place in presence of DNA dependent RNA polymerase. In some retroviruses that contain RNA as the genetic material (e.g, HIV), the flow of information is reversed. RNA synthesizes DNA by reverse transcription, then transcribed into mRNA by transcription and then into proteins by translation.

For a cell to operate, its genes must be expressed. This means that the gene products, whether proteins or RNA molecules must be made. The RNA that carries genetic information encoding a protein from genes into the cell is known as messenger RNA (mRNA). For a gene to be transcribed, the DNA which is a double helix must be pulled apart temporarily, and RNA is synthesized by RNA polymerase. This enzyme binds to DNA at the start of a gene and opens the double helix. Finally, RNA molecule is synthesized. The nucleotide sequence in the RNA is complementary to the DNA template strand from which it is synthesized.

Both the strands of DNA are not copied during transcription for two reasons. 1. If both the strands act as a template, they would code for RNA with different sequences. This in turn would code for proteins with different amino acid sequences. This would result in one segment of DNA coding for two different proteins, hence complicate the genetic information transfer machinery. 2. If two RNA molecules were produced simultaneously, double stranded RNA complementary to each other would be formed. This would prevent RNA from being translated into proteins.

### 5.9.1 Transcription unit and gene

A transcriptional unit in DNA is defined by three regions, a **promoter**, the **structural gene** and a **terminator**. The promoter is located towards the 5' end of the coding strand. It is a DNA sequence that provides binding site for RNA polymerase. The presence of promoter in a transcription unit, defines the template and coding strands. The terminator region located towards the 3' end of the coding strand contains a DNA sequence that causes the RNA polymerase to stop transcribing. In eukaryotes the promoter has AT rich regions called **TATA box** (**Goldberg-Hogness box**) and in



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